INCH-POUND
MIL-DTL-24231/11E(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/11D(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTOR, MULTIPLE CABLE, TYPE IV

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. Dimensions and configuration: see figures 1 through 11.

2. The body shall be manufactured from mill annealed and age-hardened Ni-Cu-Al alloy in accordance with QQ-N-286. The body shall be manufactured as a two-piece welded assembly.

3. After completion of machining, but prior to drilling holes for receptacles, visual and dimensional examination and hardness and dye penetrant tests for multiple cable connectors shall be performed.

4. Receptacles shall be installed in the body by the manual metal arc or tungsten inert gas welding methods and in the sequence specified in requirement 6, and as follows:

a. Clean by the vapor de-grease method or equal.

b. Clean prior to welding.

c. Size of fillet weld 3⁄16 inch. Total of three beads.

d. Liquid penetrant inspect (P.T.) root and final weld.

e. Apply weld in quadrants ¾ inch (maximum) long opposite each other until weld is complete.

f. Exercise caution during welding procedure to control temperature build-up in order to prevent any distortion of receptacles.

g. Recommended inter-pass temperature is 80 to 90 °F.

5. Where a standard gland assembly is used, any unused holes in retainer plates and packing shall be plugged by wire, type MIL-DTL-16878/36 BJE0 for 7⁄64-inch diameter holes, and LS2SWU-1 cable (see MIL-DTL-24643/33) for 17⁄64-inch diameter holes. Wires or cable shall be passed through two holes with loop on inside of fitting. Single holes shall be plugged with wire or cable with the end inside of fitting secured by a knot or other method approved by the contracting activity.

6. Two to 10 receptacles, as required, shall be installed in sequence in numbered locations shown herein. The larger receptacles shall be located in the lowest consecutive numbers.

7. Space on the designation plate (M24231/11-030) is reserved for the installing activity to stamp or engrave circuit identification directly over the respective receptacle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

8. Wires shall be secured in groups with tape beyond the end of the sleeving, according to the receptacle, and marked with position and pin numbers. Shielded cable shall also be marked with position and pin numbers at the exposed end.

9. Unless otherwise specified in the contract or order, the length of wires or cables extending from the end of the fitting shall be a minimum of 13 feet. Wires and cables shall be protected after installation by a method approved by the contracting activity.

10. The gland nut shall be tightened until the packing tightens against the wires and cables. This can be determined by moving the wires and cables gently back and forth until the wire and cable is seized by the packing. The gland nut shall be turned approximately one-fourth of a turn more to hold wires and cables in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this can damage the wires and cables.

11. Full or partial threads shall not extend more than 0.083 inch beyond the ¾-inch thread length.

12. All receptacles shall be positioned with the polarizing slot in the "DOWN" (6 o'clock) position ±2 degrees.

13. Revision C of MIL-DTL-24231/11 modified the body, M24231/11-010, for assembly with nut M24231/11-031. Existing nut, M24231/10-042 (MIL-DTL-24231/10), is retained for use as a replacement part for existing bodies, in service, manufactured to prior revisions.

14. Hull fitting body-to-cover weld may be accomplished using either of the following welding electrodes or rods in accordance with MIL-E-21562:

a. MIL-EN-60 or MIL-RN-60.

b. MIL-EN-64 or MIL-RN-64.

15. Where wire and LS2SWU-1 cable combinations differ from the table on figure 8, retainer plate, packing and retainer plate assembly shall be the same as M24231/10-081, M24231/10-082, and M24231/10-080 respectively except the number and arrangement of holes shall meet the following requirements:

a. Positional tolerancing for all holes shall be ⌖ | ∅.010.

b. Holes shall not be located closer than 0.010 from the minimum gland nut inside diameter of part number M24231/10-044 and the retainer plate position pin holes of part number M24231/10-080.

c. Minimum material between holes shall be 0.031.

d. Protective sleeving shall be sized to suit the number of wires required where use of the specified sleeving, item 4, is inadequate.

16. The outside diameter is intended to coincide with commonly available 5¼-inch bar stock. The dimension may vary after machine cleanup.

17. All dimensions are in inches.

18. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1⁄64 inch.

c. Angular dimensions – ±½ degree.

d. Tolerances for rubber packing should follow the guidance provided in table XLI (RMA Drawing Designation A2 Precision) of MIL-HDBK-149.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

2 TO 8 RECEPTACLES AS REQD M24231/11-004 SYM NO. 516
EIGHT 3-COND RECEPTACLES M24231/11-005 SYM NO. 516.1
TEN 3-COND RECEPTACLES M24231/11-006 SYM NO. 516.2
SEE REQUIREMENT 12

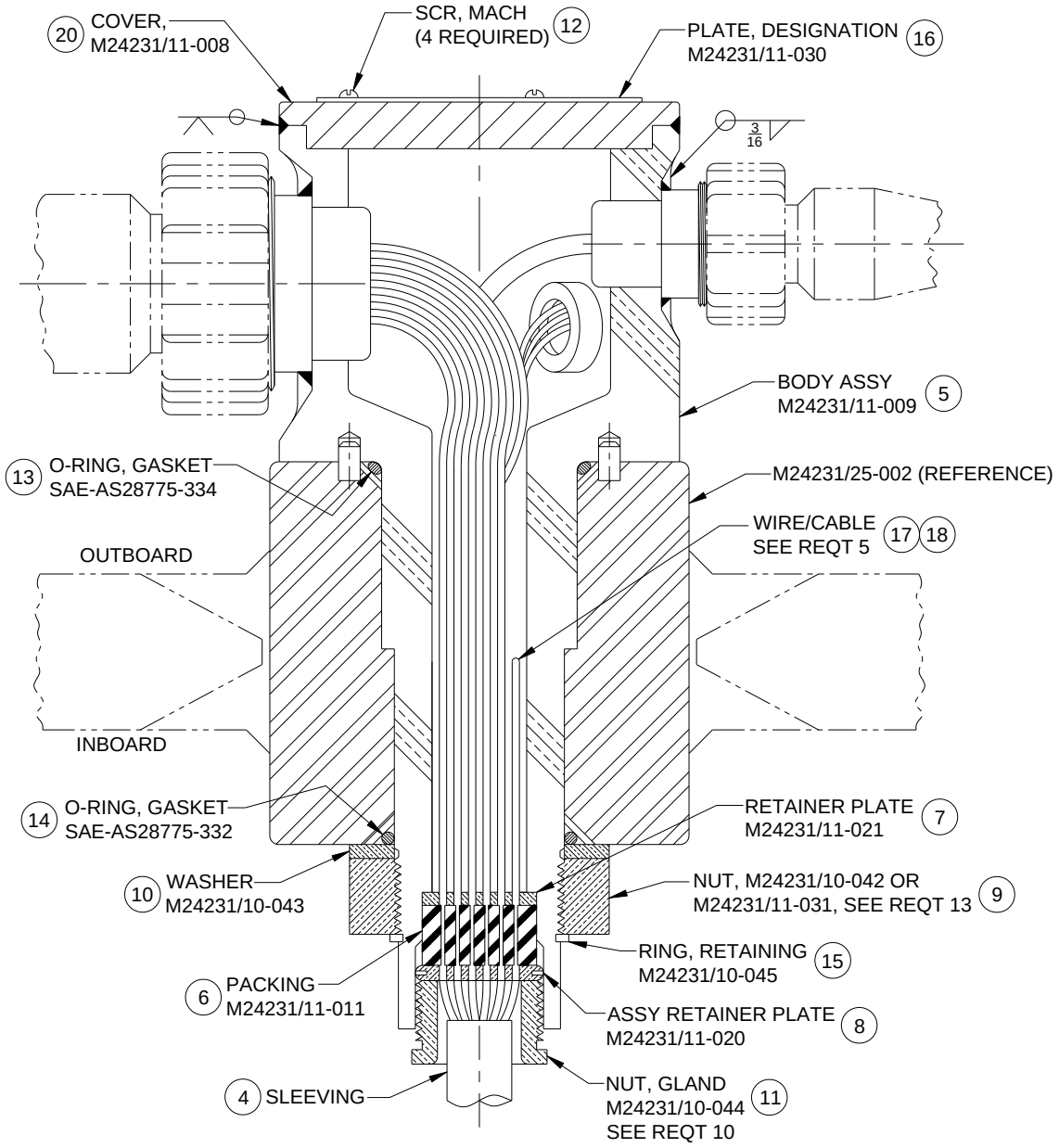


FIGURE 1. Connector, multiple cable (two-piece body). 1 2 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

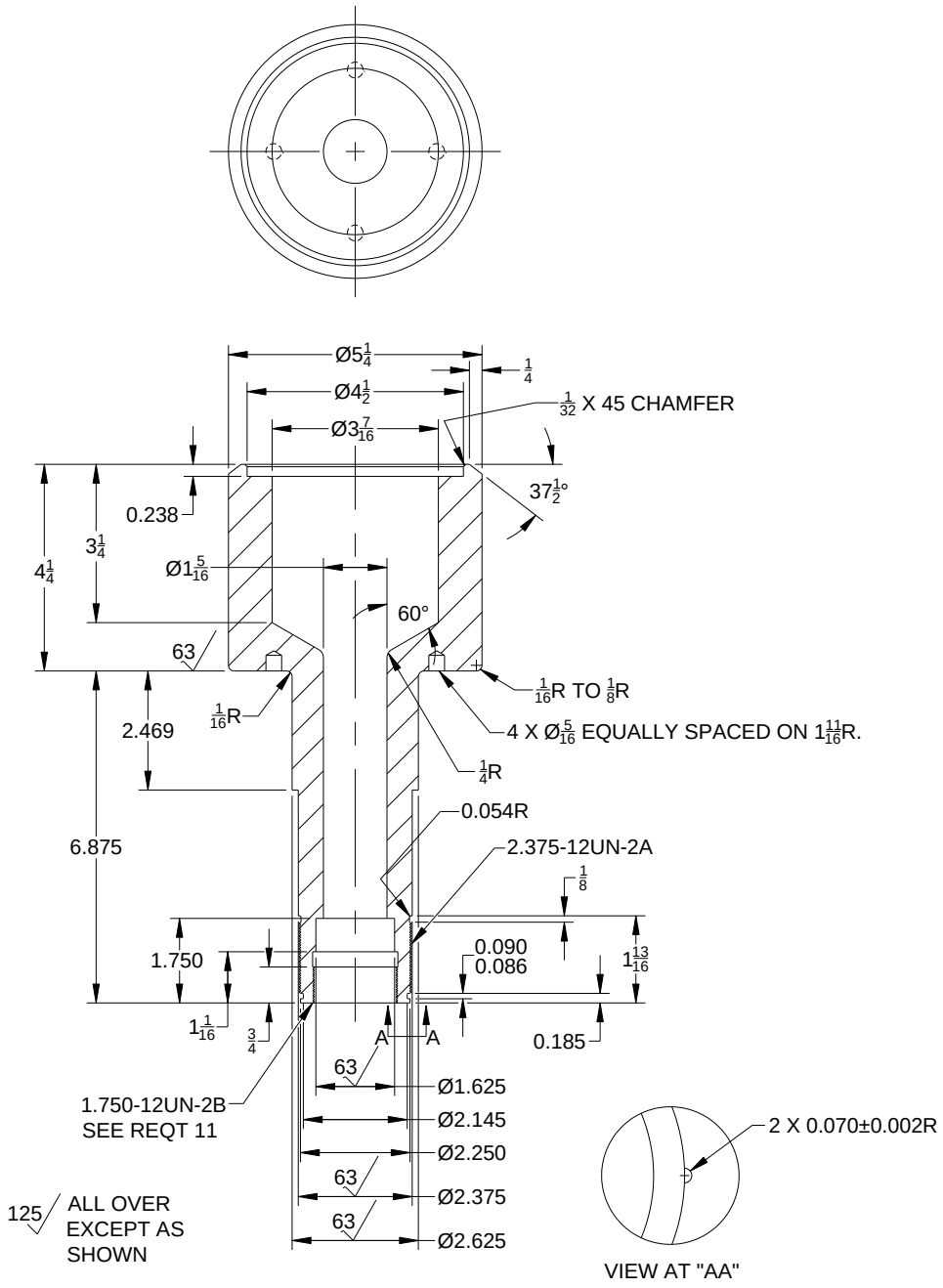


FIGURE 2. Body – M24231/11-007. (19)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

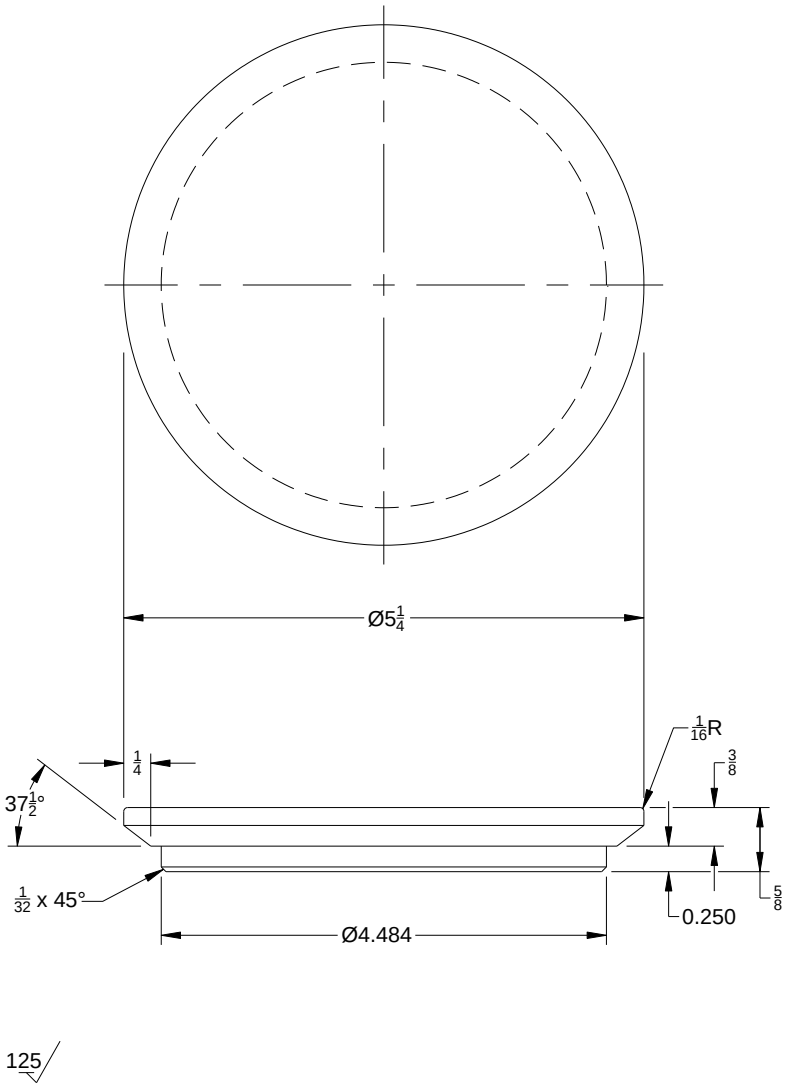
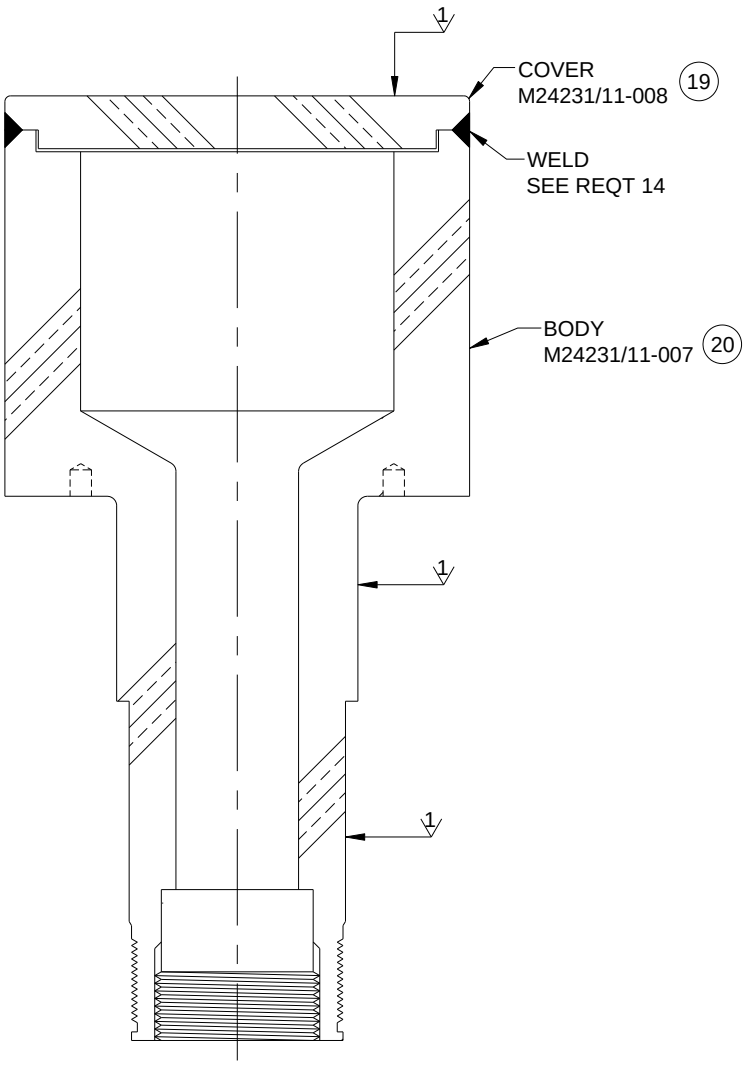


FIGURE 3. Cover – M24231/11-008. (20)

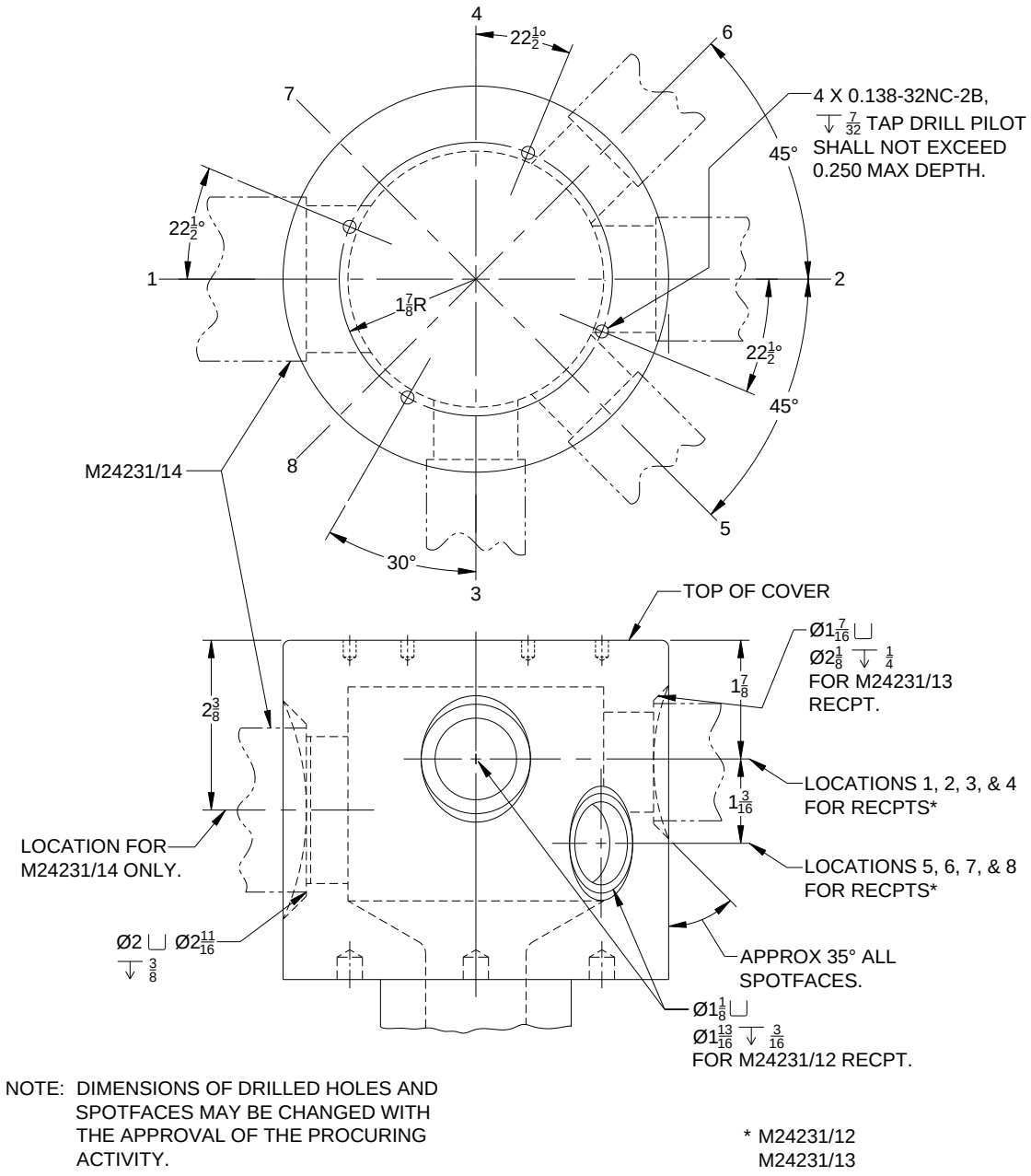
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.



1/ HARDNESS TO BE TAKEN AT THESE POINTS.

FIGURE 4. Body assembly – M24231/11-009. 5

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.



NOTE: DIMENSIONS OF DRILLED HOLES AND SPOTFACES MAY BE CHANGED WITH THE APPROVAL OF THE PROCURING ACTIVITY.

* M24231/12
M24231/13

FIGURE 5. Typical location of receptacles and drilling sequence for M24231/11-004 (see requirement 6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

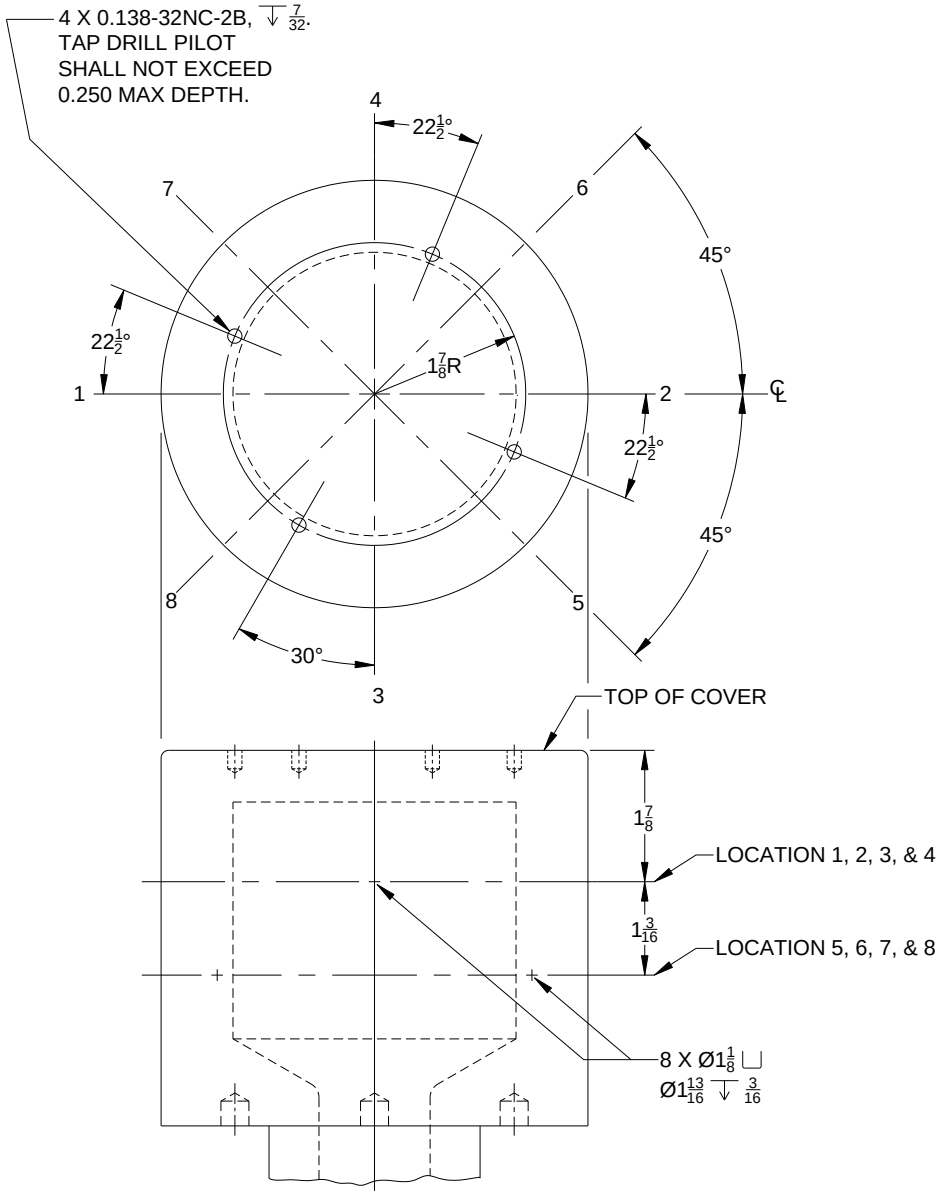


FIGURE 6. Location of receptacles for M24231/11-005.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

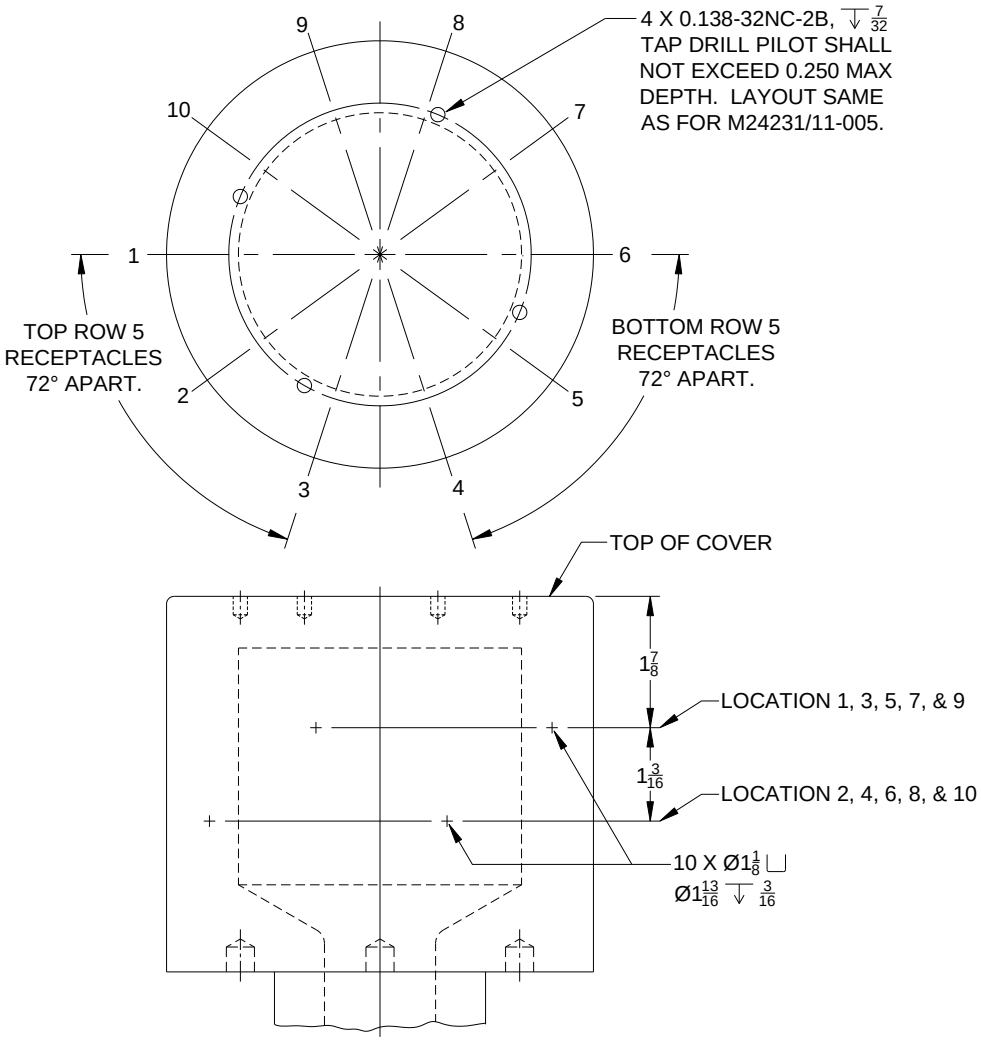


FIGURE 7. Location of receptacles for M24231/11-006.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

M24231/11-021 SAME AS M24231/10-081 EXCEPT HOLE SIZES AND LOCATIONS AS SHOWN.
M24231/11-020 SAME AS M24231/10-080 EXCEPT HOLE SIZES AND LOCATIONS AS SHOWN.
M24231/11-011 SAME AS M24231/10-082 EXCEPT HOLE SIZES AND LOCATIONS AS SHOWN.

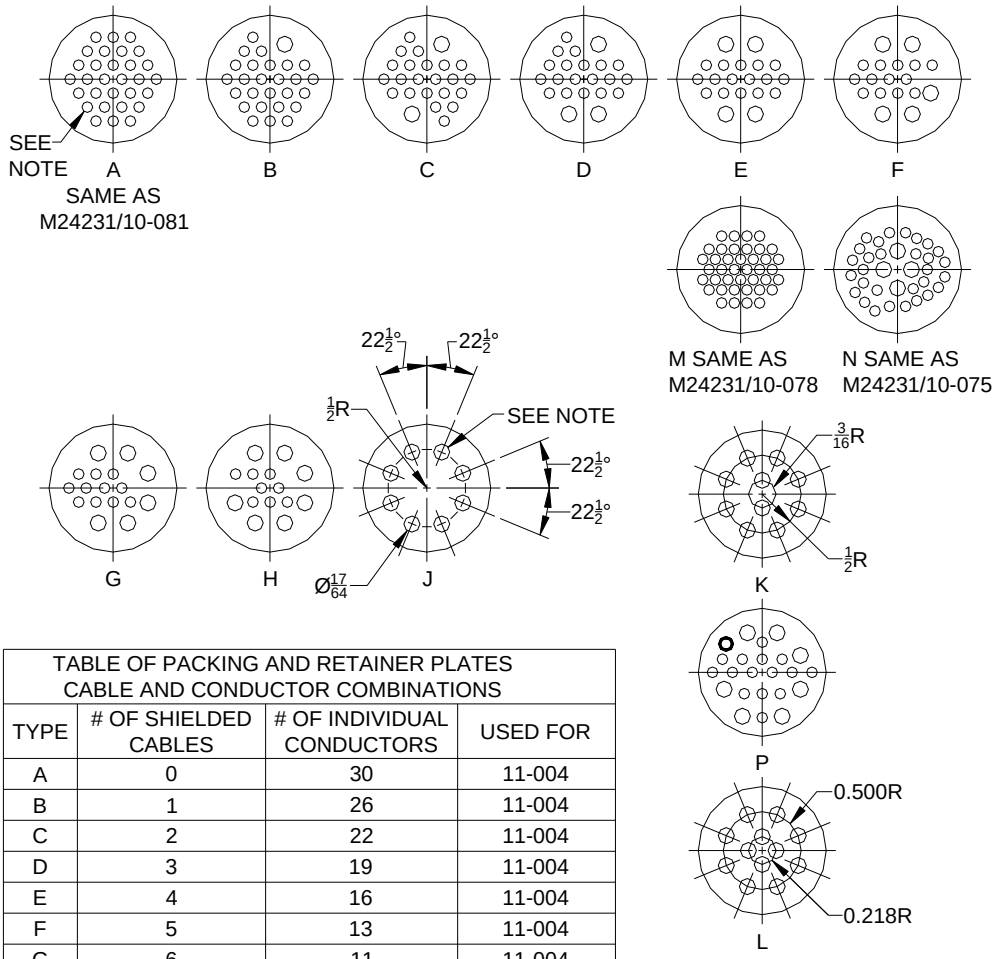


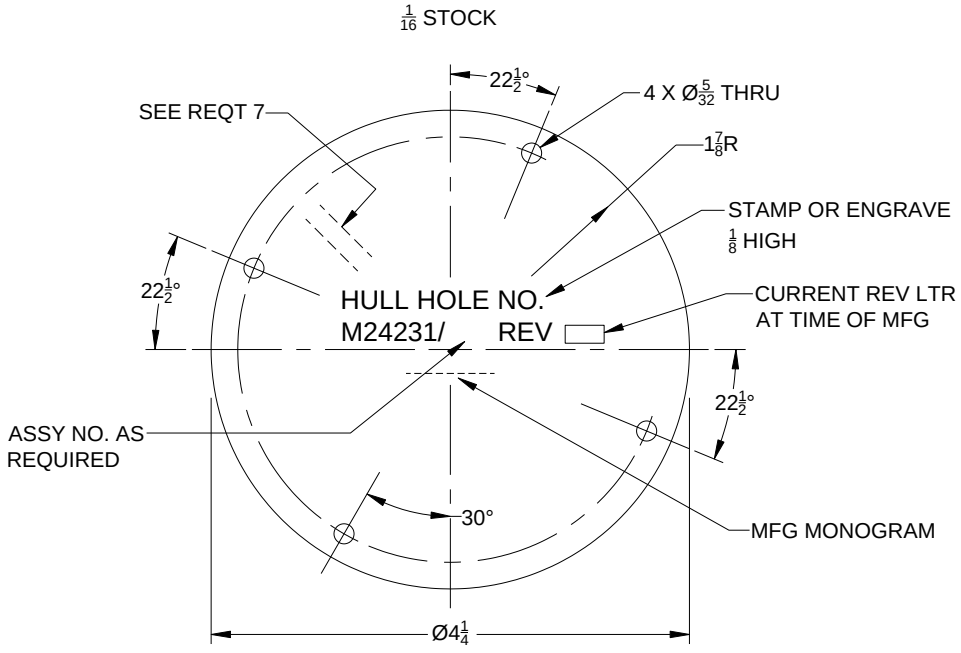
TABLE OF PACKING AND RETAINER PLATES CABLE AND CONDUCTOR COMBINATIONS			
TYPE	# OF SHIELDED CABLES	# OF INDIVIDUAL CONDUCTORS	USED FOR
A	0	30	11-004
B	1	26	11-004
C	2	22	11-004
D	3	19	11-004
E	4	16	11-004
F	5	13	11-004
G	6	11	11-004
H	7	8	11-004
J	8	0	11-004, 11-005
K	10	0	11-006
L	12	0	
M	0	40	
N	4	30	
P	7	13	11-004

SEE REQUIREMENT 15 FOR GENERAL INFORMATION ON PLATE AND PACKING HOLE NUMBER AND ARRANGEMENT.

NOTE: THE NUMBER OF 7/64 AND 17/64 DIA HOLES TO VARY FOR EACH TYPE AS SHOWN. LOCATIONS OF 7/64 DIA HOLES SHALL BE SAME AS TYPE "A" AND THE 17/64 DIA HOLES THE SAME AS TYPE "J" (EXCEPT TYPES "M" AND "N").

FIGURE 8. Retainer plates and packings. (6)(7)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.



63

NOTE: DIMENSIONS AND SPOTFACES MAY BE CHANGED WITH THE APPROVAL OF THE CONTRACTING ACTIVITY.

FIGURE 9. Designation plate – M24231/11-030. (16)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

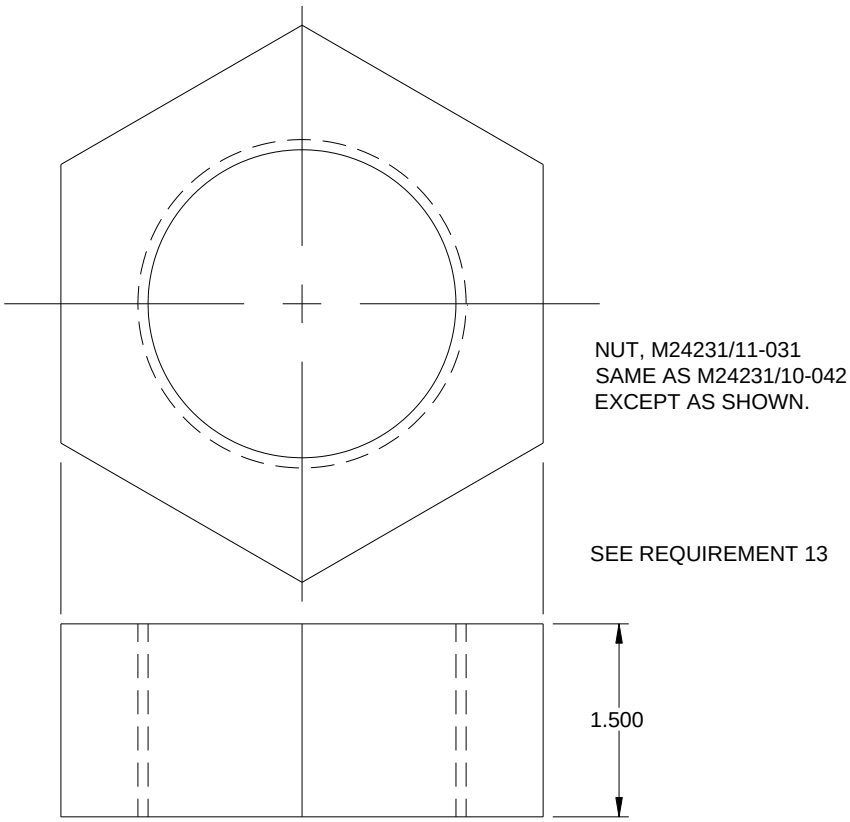
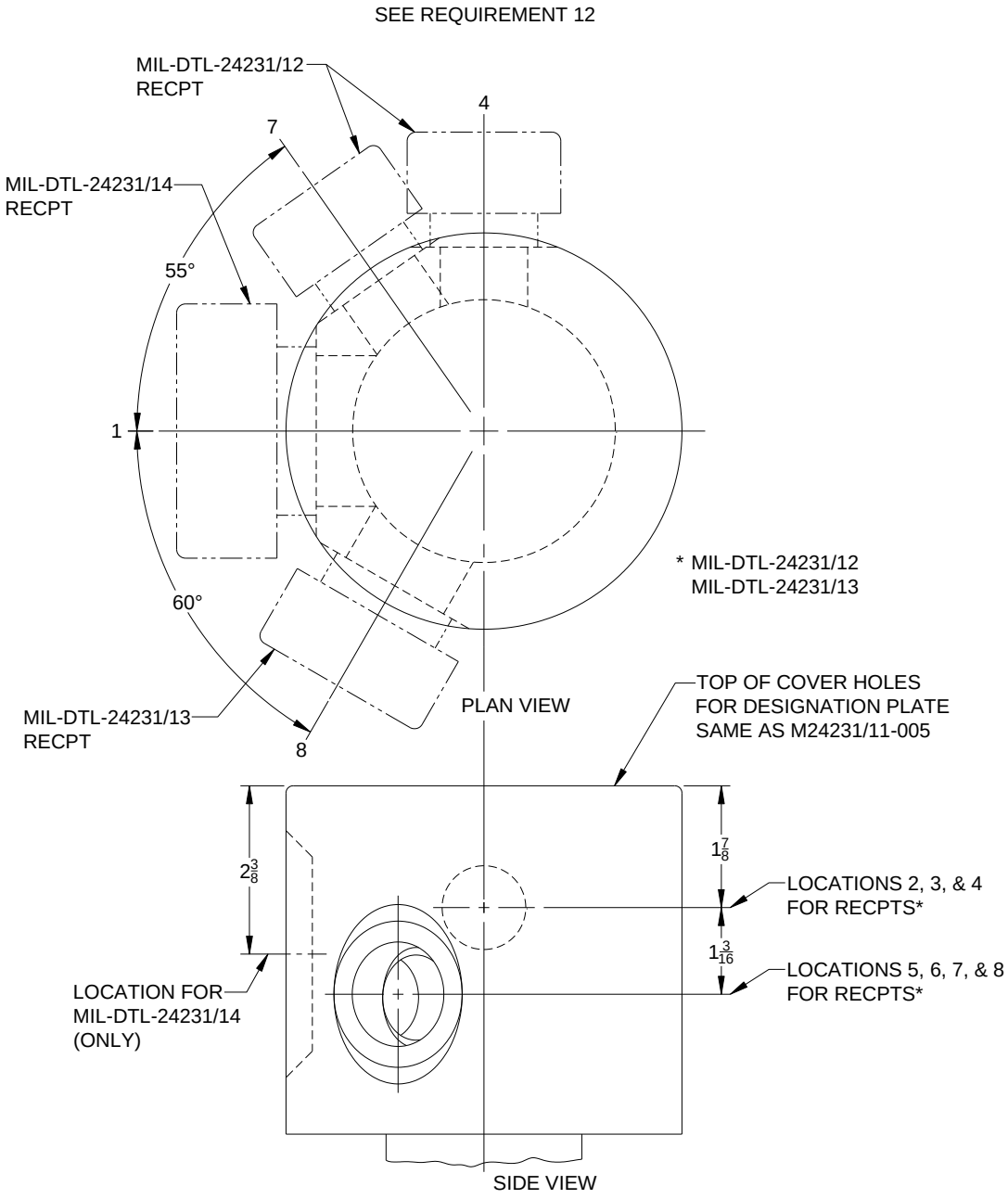


FIGURE 10. Nut – M24231/11-031. (9)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.



NOTE: THE RECEPTACLE SIZES AND THEIR LOCATIONS AS SHOWN ARE SOMETIMES REQUIRED DUE TO SYSTEM CHANGES AND SHIPBOARD INTERFERENCES. THE LOCATING DEGREES INDICATED SHALL BE USED IN ORDER TO AVOID THE INTERSECTION OF HOLES AND SPOTFACES.

FIGURE 11. Location of receptacles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

TABLE I. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/11- 3/			Material/specification	Remarks
		004	005	006		
1	Assembly, multiple cable, connector, symbol no. 516	1			-----	2-8 receptacles; see figure 1.
2	Assembly, multiple cable, connector, symbol no. 516.1		1		-----	(8) – 3-conductor receptacles; see figure 1.
3	Assembly, multiple cable, connector, symbol no. 516.2			1	-----	(10) – 3-conductor receptacles; see figure 1.
4	Sleeving, 0.890 id, 0.050 thick, 10 ft long	1	1	1	Type F, Form U, Category 1, color clear, MIL-I-631	See figure 1.
5	Assembly, two-piece body	1	1	1	See items 19 and 20.	See figure 2, figure 3, and figure 4.
6	Class A, packing durometer 55±5	1	1	1	Rubber, ZZ-R-768	See figure 1 and figure 8, type as required.
7	Plate, retainer	1	1	1	Ni-Cu, Class A, QQ-N-281	See figure 1 and figure 8, type as required.
8	Assembly, retainer plate	1	1	1	Ni-Cu, Class A, QQ-N-281	See figure 1 and figure 8, type as required.
9	Nut	1	1	1	Ni-Cu-Al, QQ-N-286	See figure 1 and figure 10, requirement 13.
10	Washer	1	1	1	M24231/10-043	See figure 1.
11	Nut, gland	1	1	1	M24231/10-044	See figure 1.
12	Screw, machine, 0.138-32NC-2A by 3/16 long	4	4	4	CRES, coml	See figure 1.
13	Gasket, O-ring	1	1	1	SAE-AS28775-334 4/	Lubricate with silicone compound, SAE-AS8660; see figure 1.
14	Gasket, O-ring	1	1	1	SAE-AS28775-332 4/	See item 13.
15	Ring, retaining	1	1	1	M24231/10-045	See figure 1.
16	Plate, designation	1	1	1	CRES or Al, MIL-DTL-15024	See figure 1 and figure 9.
17	Wire, B-16	As required			Nylon jacket, SAE-AS22759	See figure 1, requirements 5 and 15.
18	Cable, LS2SWU-1	As required			MIL-DTL-24643/33	See figure 1, requirements 5 and 15.
19	Body	1	1	1	Ni-Cu-Al, QQ-N-286	Part of assembly, two-piece body; see item 5.
20	Cover, body	1	1	1	Ni-Cu-Al, QQ-N-286	Part of assembly, two-piece body, see item 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.

TABLE I. Parts list – Continued.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Nitrile (Buna N); Parker Hannifin Corporation (part number N0756-75), O-ring Division, 2360 Palumbo Drive, Lexington, KY 40509, CAGE 02697, or equal.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-060)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:14Z
Check the source to verify that this is the current version before use.